BIG DATA ANALYTICS

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DEFINITION OF BIG DATA

- Big data refers to datasets that are not only big, but also high in variety and velocity, which makes them difficult to handle using traditional tools and techniques
  - (Nada Elgendy and Ahmed Elragal.2014.Big Data Analytics: A Literature Review Paper)
- Big Data refers to large sets of complex data, both structured and unstructured which traditional processing techniques and/or algorithms are unable to operate on.
  - (Kevin Taylor-Sakyi.2016.Big Data: Understanding Big Data)
- Signifcant volumes of stored data or data regarding electronic transactions that are used in support of decision making, with managers, policymakers, and executive officers now routinely embracing technology to transform these abundant raw data into useful, informative information.
  - (Renu Sabharwal and Shah Jahan Miah.2021.A new theoretical understanding of big data analytics capabilities in organizations: a thematic analysis.)

DEFINITION OF BIG DATA

- Big Data Analytics" (BDA)—the application of advanced analytic techniques, including data mining, statistical analysis, and predictive modeling on big datasets as new business intelligence practice.
- BDA uses computational intelligence techniques to transform raw data into information that can be used to support decision-making.

Term (Abbreviation)	Approximate Size
Byte (B)	8 bits
Kilobyte (KB)	1024 bytes / 10^3 bytes
Megabyte (MB)	1024 KB / 10^6 bytes
Gigabyte (GB)	1024 MB / 10^9 bytes
Terabyte (TB)	1024 GB / 10^{12} bytes
Petabyte (PB)	1024 TB / 10^{15} bytes
Exabyte (EB)	1024 PB / 10^{18} bytes
Zettabyte (ZB)	1024 EB / 10^{21} bytes
Yottabyte (YB)	1024 ZB / 10^{24} bytes

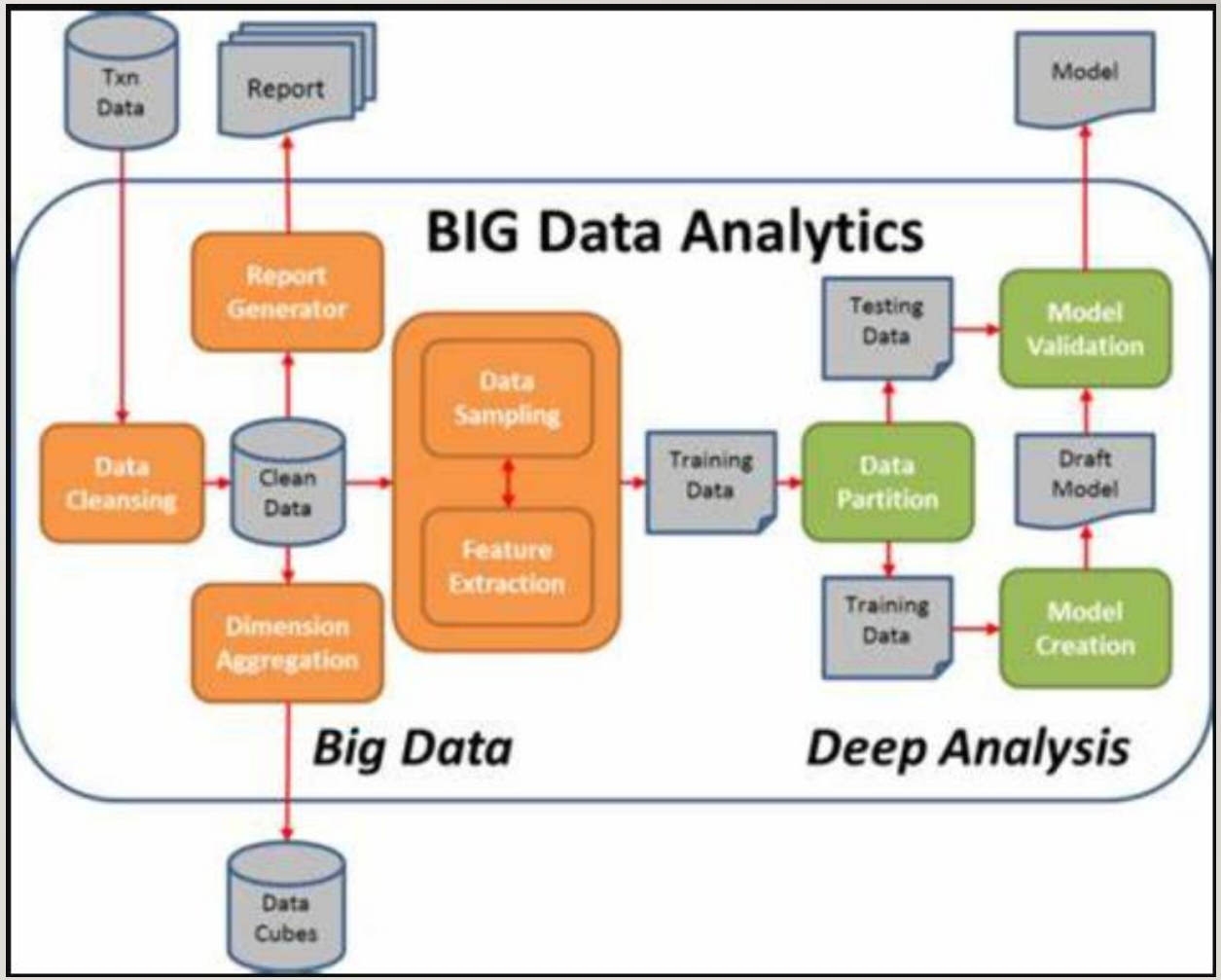


BIG DATA ISSUES

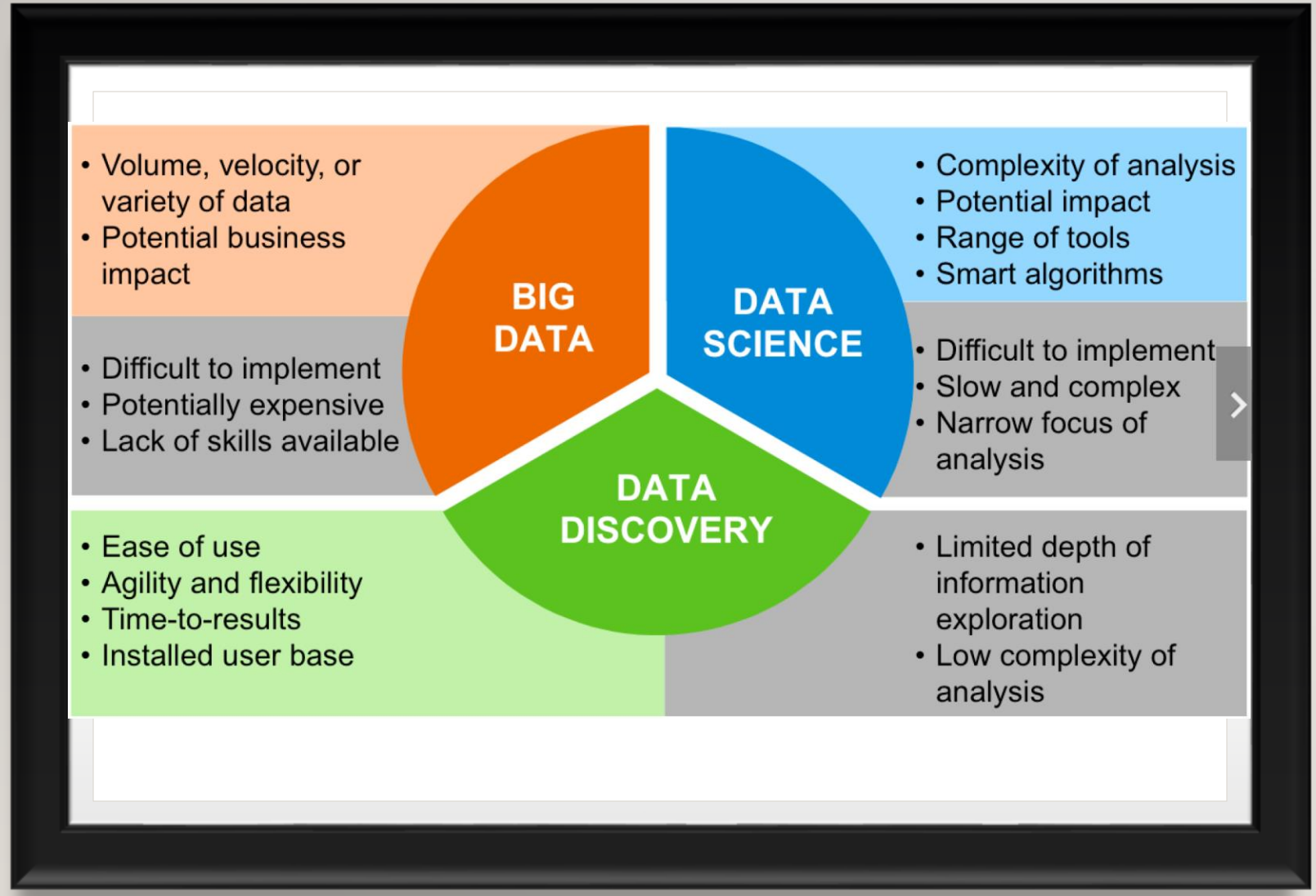
Conceptual of Big data

- Volume – Current data existing is in petabytes, which is already problematic; it's predicted that in the next few years it's to increase to zettabytes (ZB). This is due to increase use of mobile devices and social networks primarily.
- Velocity – Refers to both the rate at which data is captured and the rate of data flow. Increased dependability on live data cause challenges for traditional analytics as the data is too large and continuously in motion.
- Variety – As data collected is not of a specific category or from a single source, there are numerous raw data formats, obtained from the web, texts, sensors, e-mails, etc. which are structured or unstructured. This large amount causes old traditional analytical methods to fail in managing big data.
- Veracity – Ambiguity within data is the primary focus in this dimension – typically from noise and abnormalities within the data
- Valuable - The usefulness or benefit derived from the data. Data by itself is not enough—what matters is the insight, decisions, and actionable outcomes that come from processing and analyzing it.

BIG DATA CONCEPT



BIG DATA COVERAGE



BIG DATA ANALYTICS - BDA

- Big data analytics describes the process of uncovering trends, patterns, and correlations in large amounts of raw data to help make data-informed decisions.
  - collecting, processing, cleaning, and analyzing large datasets ➔ to help organizations operationalize their big data.
- Datasets → A data set is an ordered collection of data

CHALLENGES OF BIG DATA

- **Making big data accessible.** Collecting and processing data becomes more difficult as the amount of data grows. Organizations must make data easy and convenient for data owners of all skill levels to use.
- **Maintaining quality data.** With so much data to maintain, organizations are spending more time than ever before scrubbing for duplicates, errors, absences, conflicts, and inconsistencies.
- **Keeping data secure.** As the amount of data grows, so do privacy and security concerns. Organizations will need to strive for compliance and put tight data processes in place before they take advantage of big data.
- **Finding the right tools and platforms.** New technologies for processing and analyzing big data are developed all the time. Organizations must find the right technology to work within their established ecosystems and address their particular needs. Often, the right solution is also a flexible solution that can accommodate future infrastructure changes.

TASK 1.

- Review 4-5 paper which related with Big Data concept
- find an example of a dataset and explain that the dataset can be processed in the big data concept.
- Group @ 3-4 student.

THANK YOU!

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